

In the news...

'Happy evolutionary accident': How crayfish living underground may have gotten their bright colors



A new study suggests the trait may have happened randomly and not as a survival tactic. In a study published in July 2024, in the *Proceedings of the Royal Society B*, an ecologist at West Liberty University, and colleagues propose that the colorful hues of crayfish evolved not for a specific purpose—such as to attract mates or warn predators—but instead purely by chance.

What Makes Your Brain Different From a Neanderthal's?



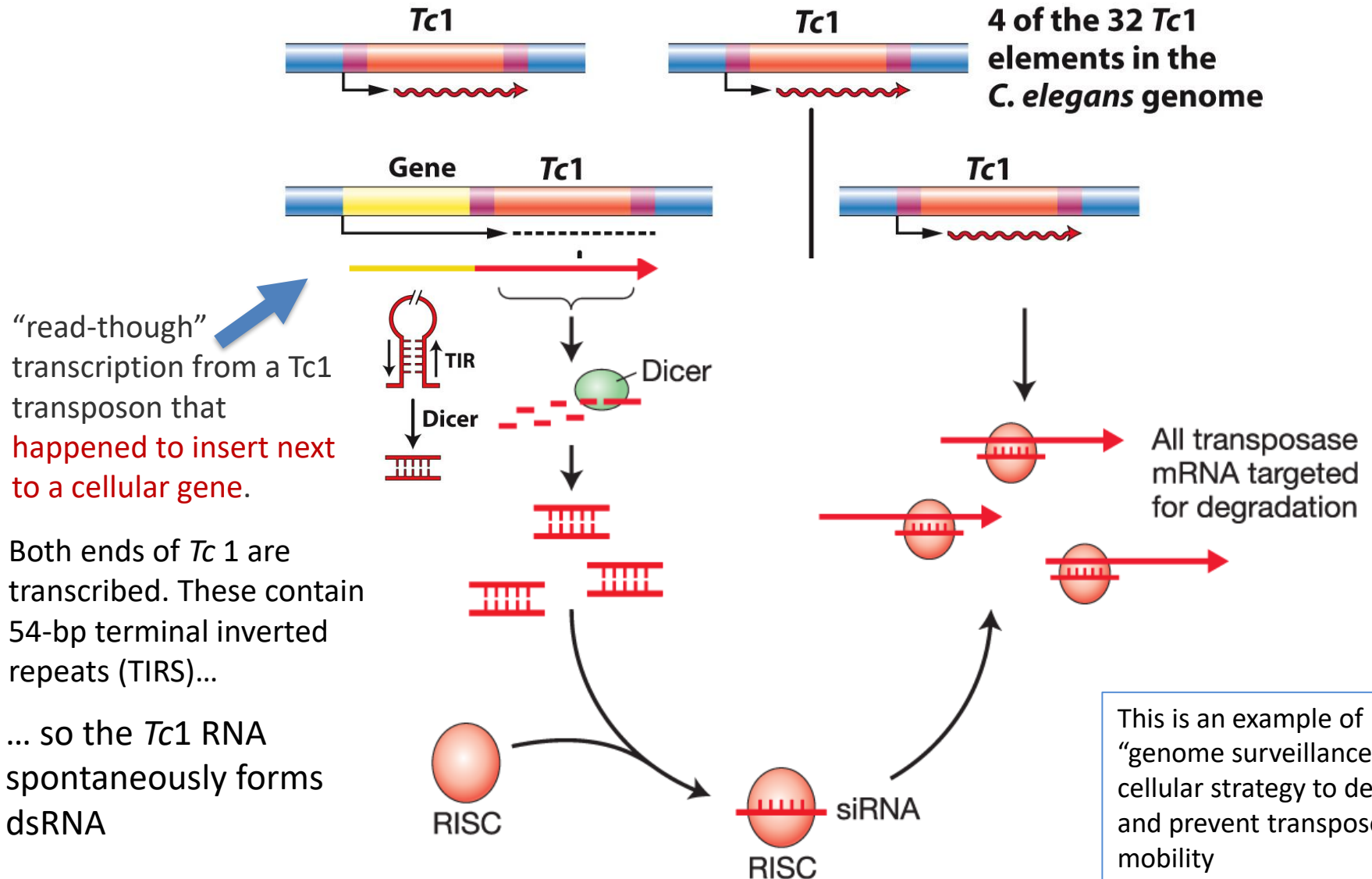
Paleoanthropologists have found that the brains of our ancestors increased in size starting about two million years ago.

Scientists have discovered a mutation that increases the production of brain cells and seems to have set our ancestors apart from other hominins.

The Dynamic Genome

- Discovery of transposable elements through study of “unstable” alleles
- Impact of transposable elements on the genome
- Genomic surveillance mechanisms that limit transposon mobility

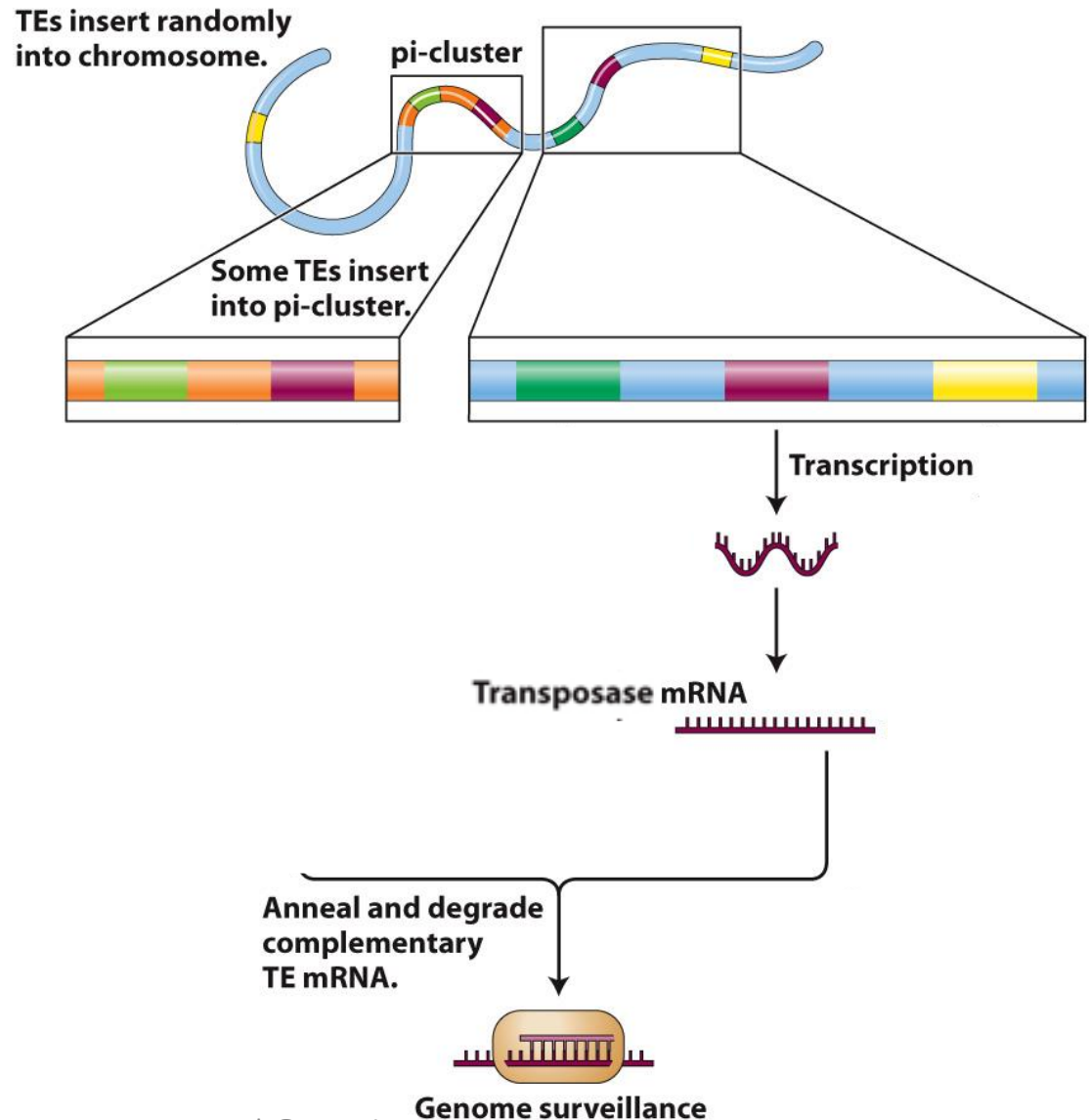
A single Tc1 can repress all Tc1 mobility in a cell



piRNAs repress transposon mobility in the germ lines of *Drosophila* and other animals

Like siRNAs, piRNAs are short single-stranded RNAs (26–30 nt in mammals) that interact with a protein complex that destroys complementary target mRNAs.

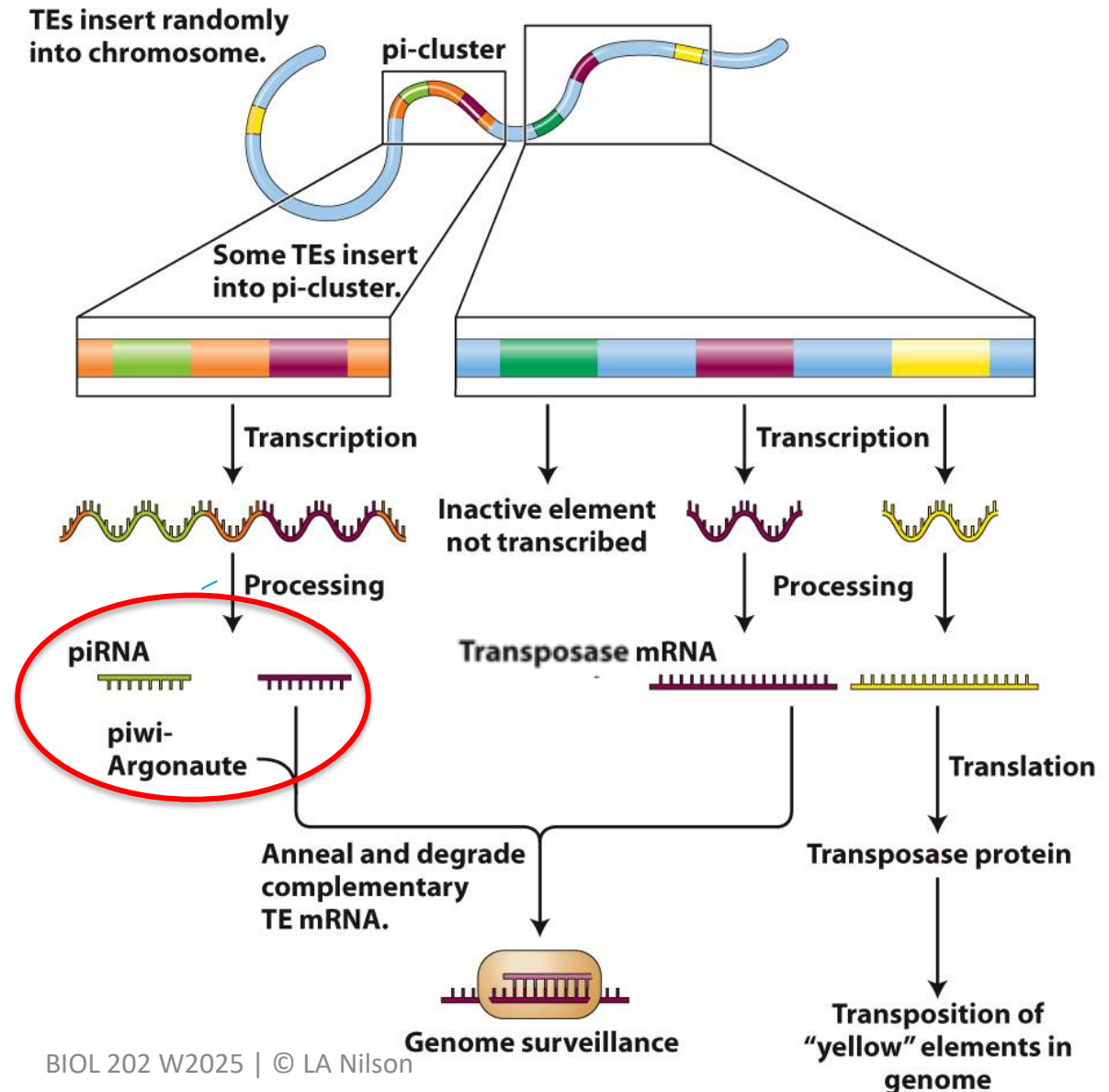
Unlike siRNAs, piRNAs do not start out as double-stranded RNA, but instead as parts of long mRNAs from a specific genome region called a pi-cluster.



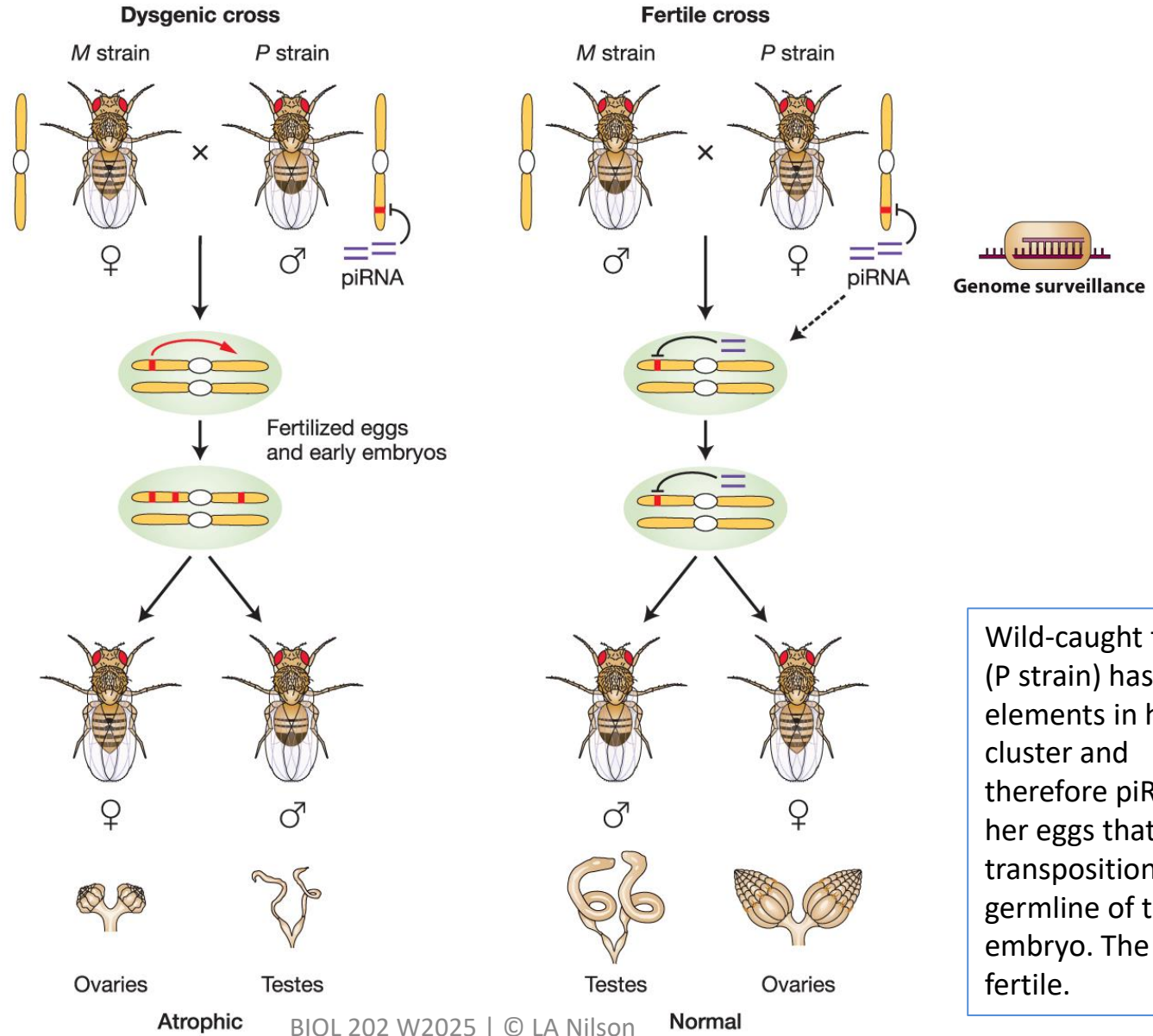
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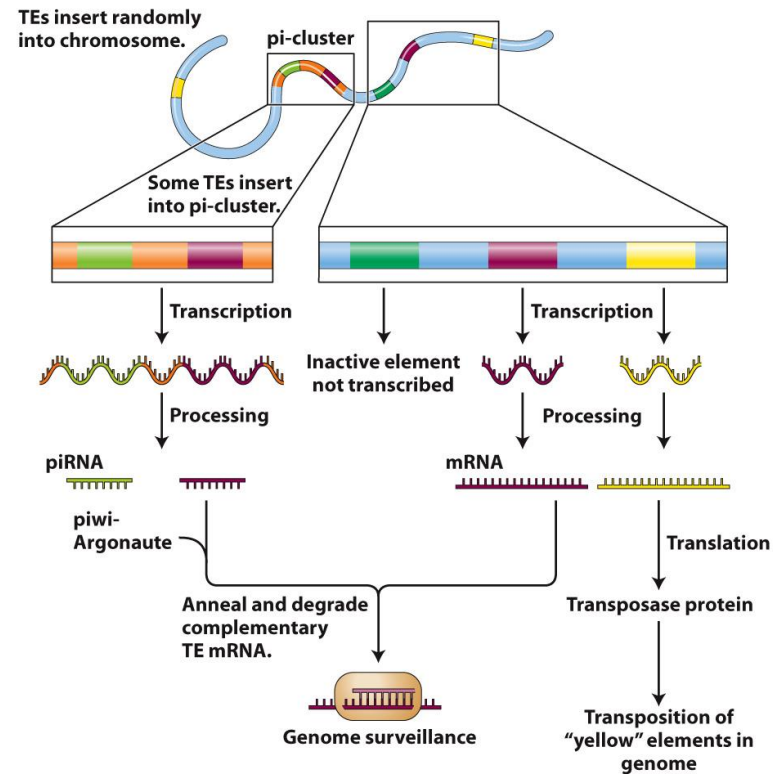
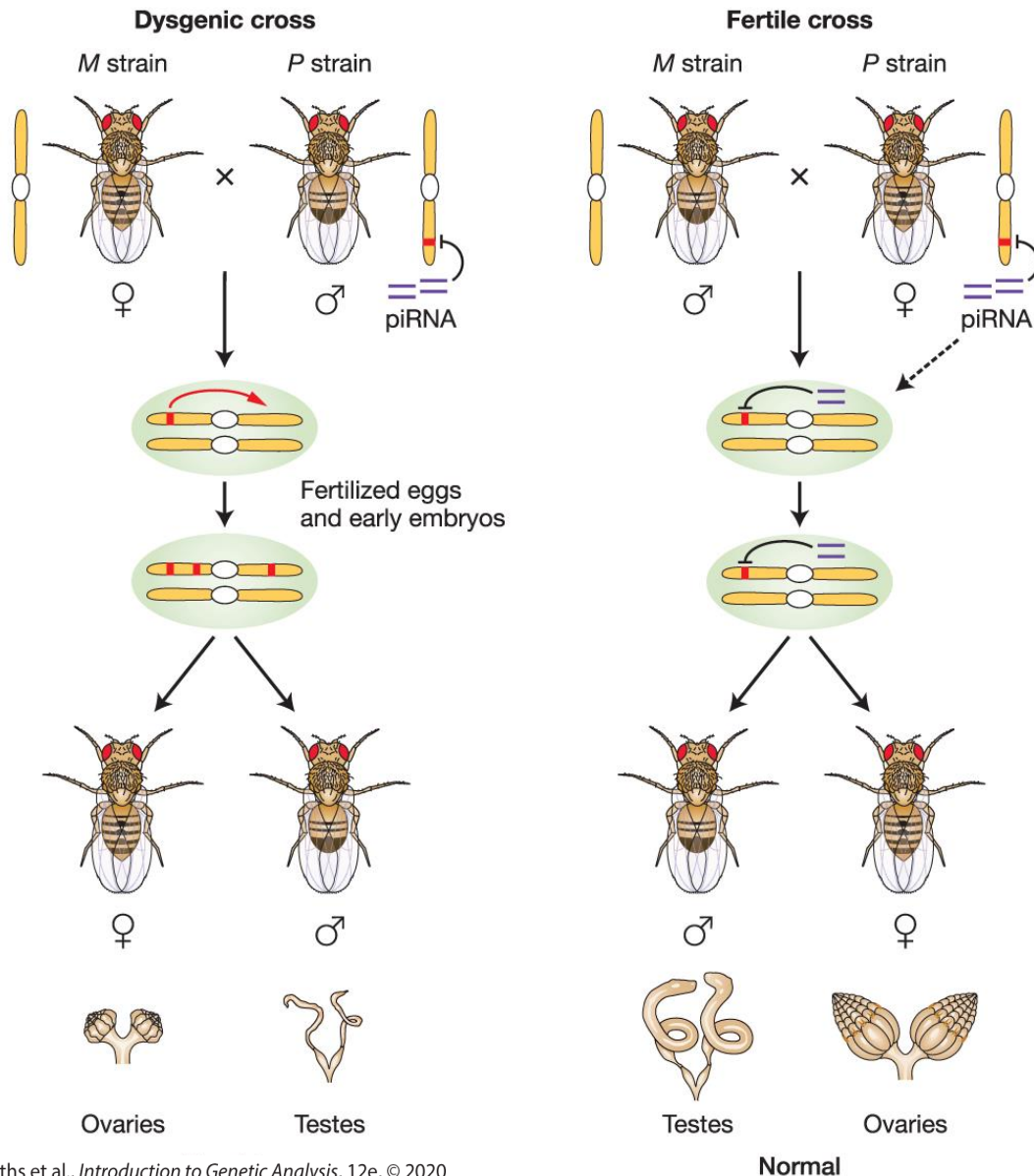
“Hybrid dysgenesis” explained: piRNAs are present in P strain but not M strain



Lab strain female (M strain) has no P elements in her pi cluster and therefore no piRNAs in her eggs. P elements introduced in the male genome therefore can mobilize, and this causes genome instability in the F1 germline, causing atrophic gonads and sterility.

Wild-caught female (P strain) has P elements in her pi cluster and therefore piRNAs in her eggs that block transposition in the germline of the embryo. The F1 are fertile.

“Hybrid dysgenesis” explained: piRNAs are present in P strain but not M strain

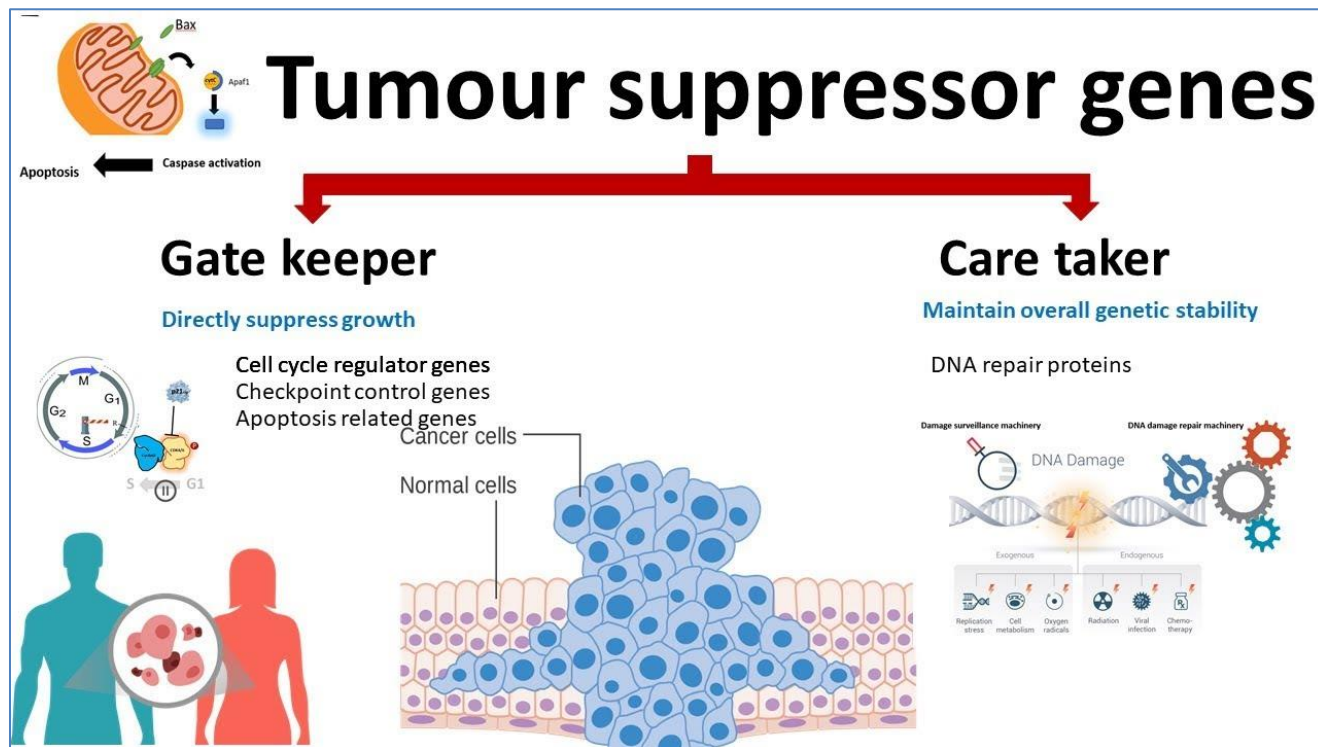


Repression of transposable element mobility in humans may suppress tumor formation

The p53 gene is one of the most studied mutations in human cancer.

Mutations in p53 are found in most tumor types and contribute to the complex network of molecular events leading to tumor formation.

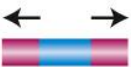
Hypothesis: p53 plays a role in repressing transposable element mobility.

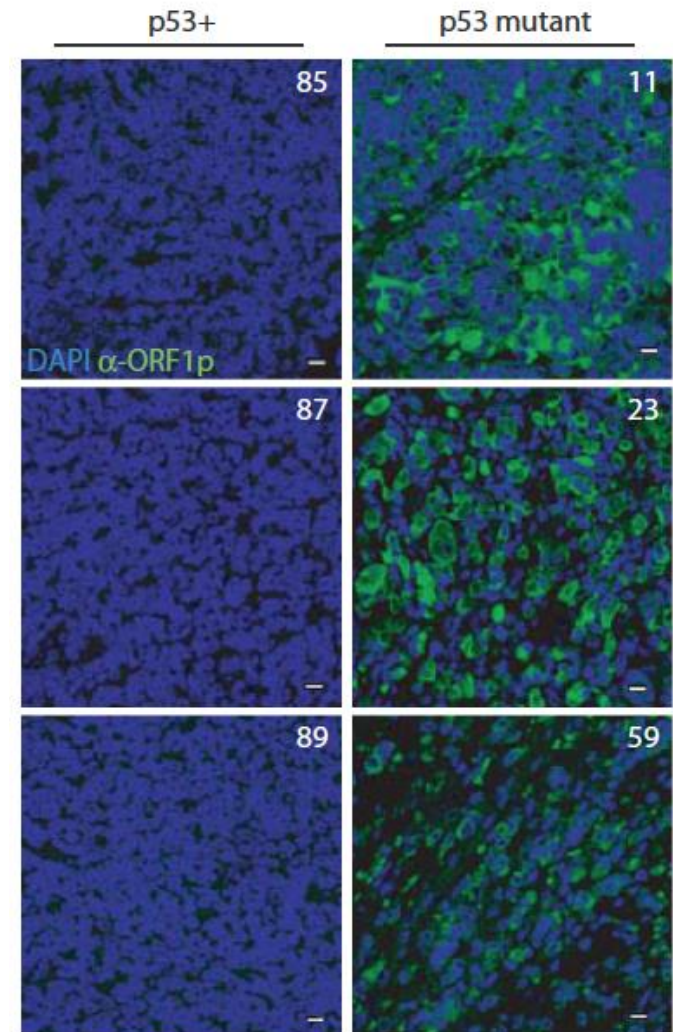


Wilms (kidney) tumors mutant for p53 show elevated human LINE1 ORF1p expression

blue = DNA

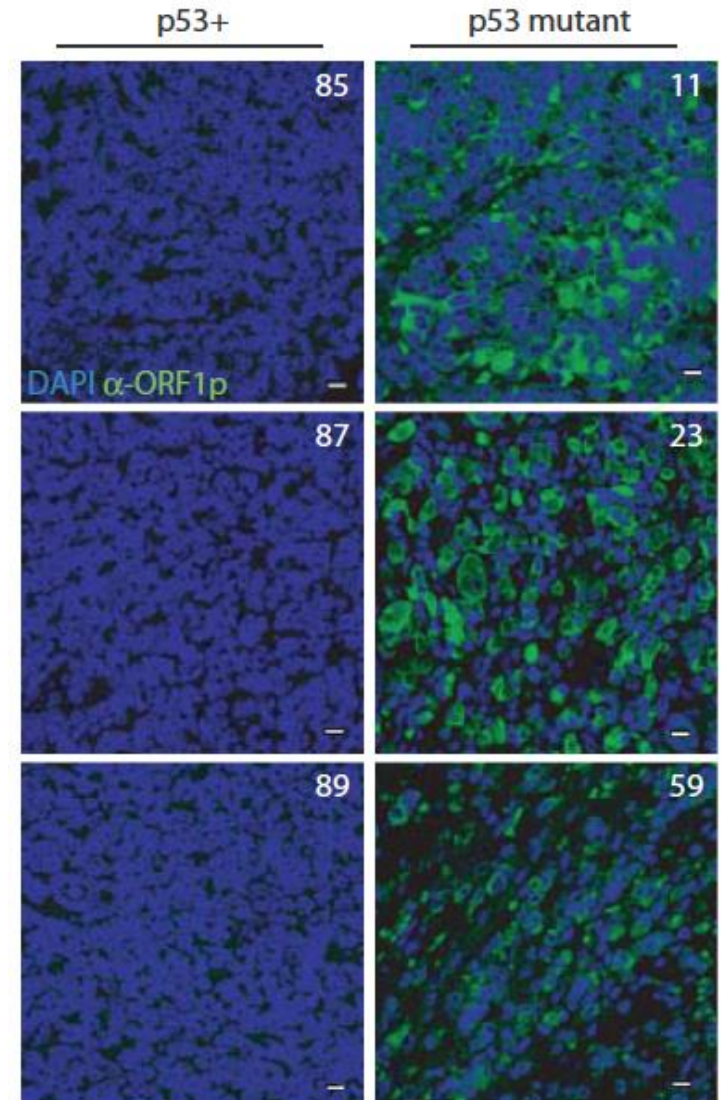
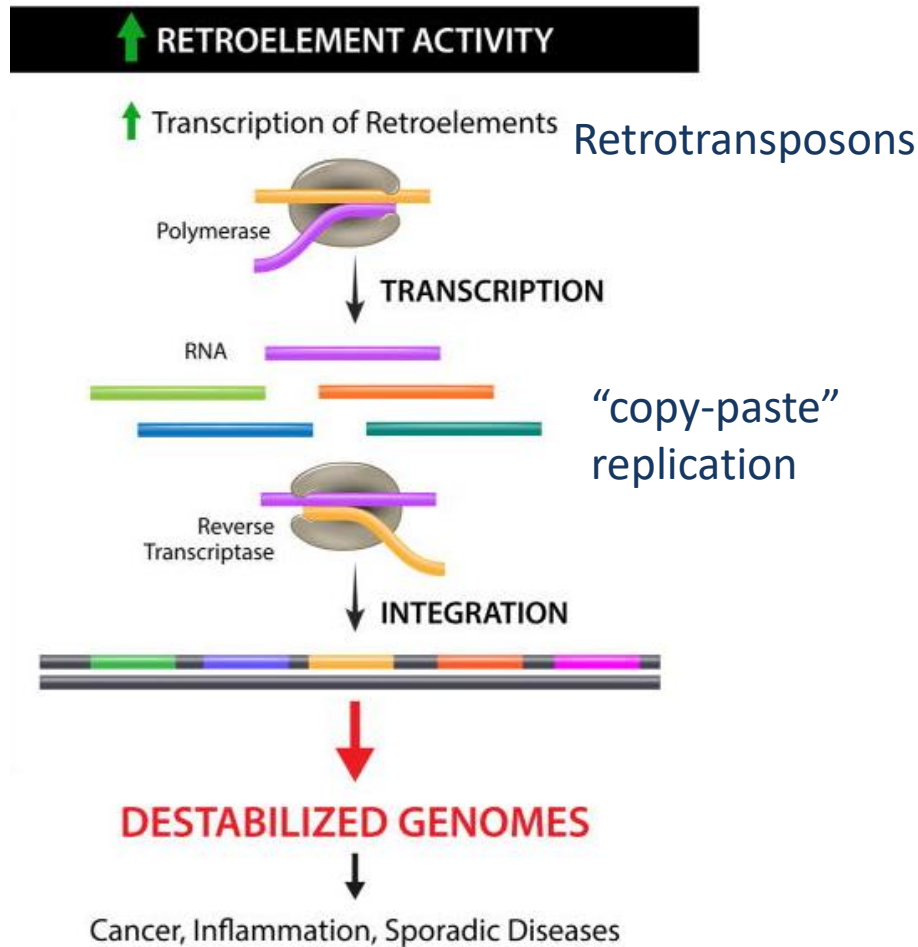
green = human LINE-1 ORF1p expression

Element	Transposition	Structure
LINES	Autonomous	
SINEs	Nonautonomous	
DNA transposons	Autonomous	
	Nonautonomous	

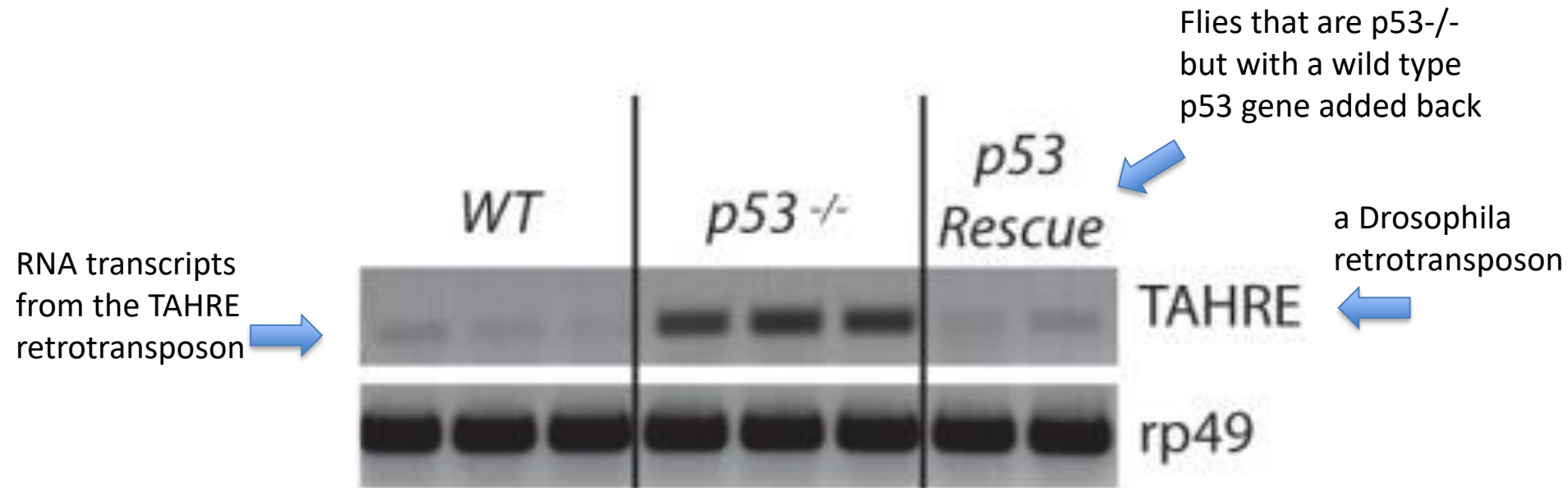


Deregulated retroelements stratify with p53 mutations in Wilms tumors. Compared with Wilms tumors that are wild type for p53 (*left* panels: 85, 87, and 89), Wilms tumors that are mutant for p53 (*right* panels: 11, 23, and 59) show dramatically elevated human LINE-1 ORF1p expression (α -ORF1p [green]; counterstained to visualize DNA [blue])

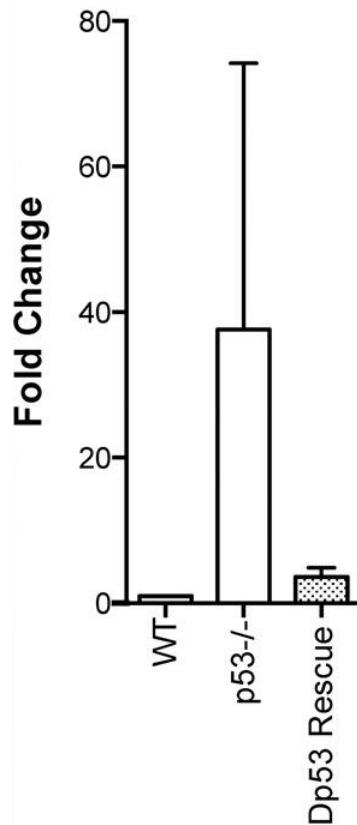
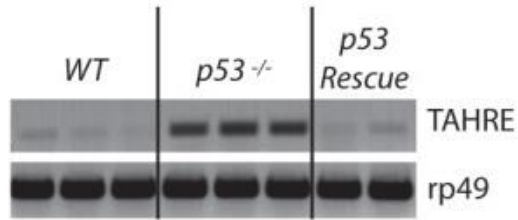
Hypothesis: Loss of p53 in kidney cells leads to deregulation of retrotransposon activity and to tumorigenesis.



The p53 protein is required for silencing transposon mobility in *Drosophila*

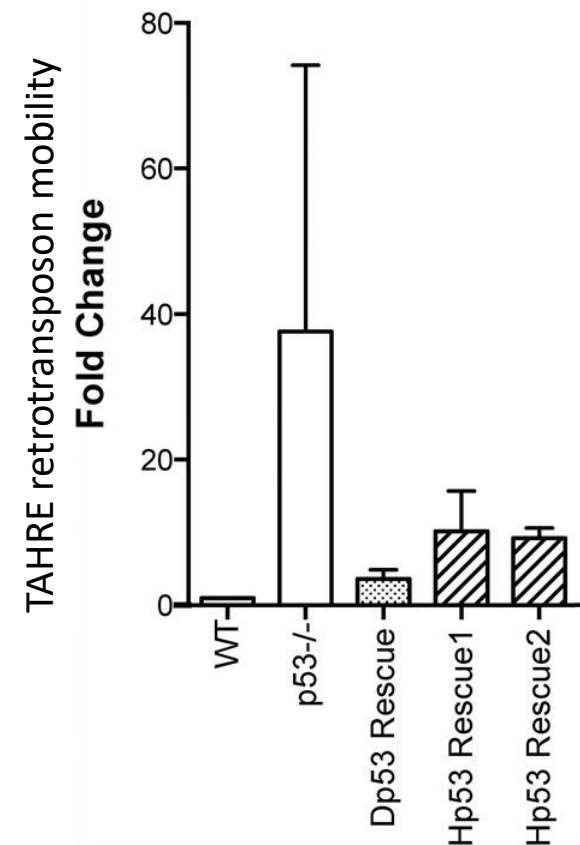
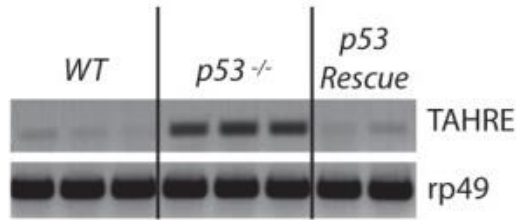


The p53 protein is required for silencing transposon mobility in *Drosophila*



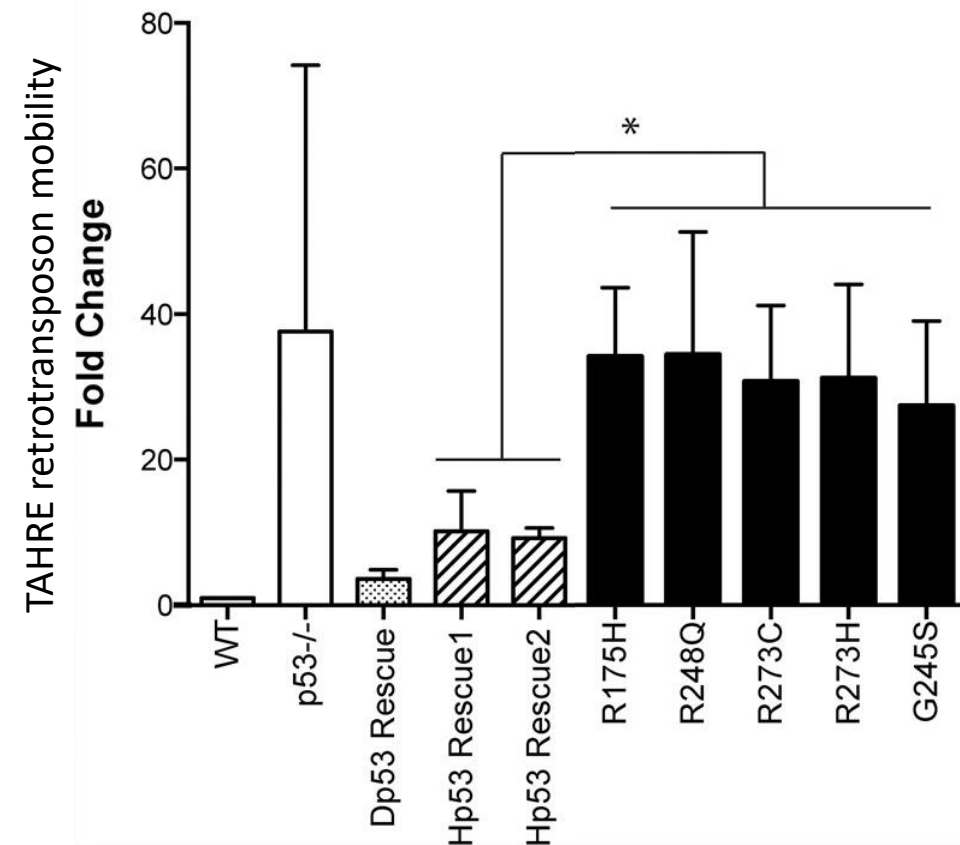
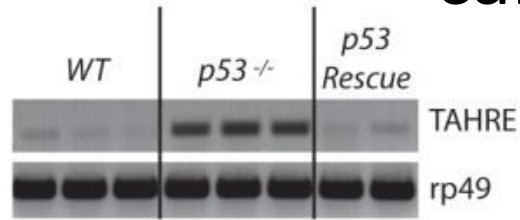
1. WT : wild type fly
2. p53^{-/-} : P53 mutant fly
3. Dp53 Rescue: p53 mutant fly + transgenic wild type *Drosophila* p53

Human p53 can block retrotransposon activity in p53-mutant flies



1. WT : wild type fly
2. p53^{-/-} : P53 mutant fly
3. Dp53 Rescue: p53 mutant fly + transgenic wild type Drosophila p53
4. Hp53 Rescue1: p53 mutant fly + transgenic wild type human p53
5. Hp53 Rescue2: p53 mutant fly + transgenic wild type human p53 (replicate)

Human p53 corrects dysregulated transposon activity in p53^{-/-} flies, but p53 variants commonly seen in cancer patients do not.

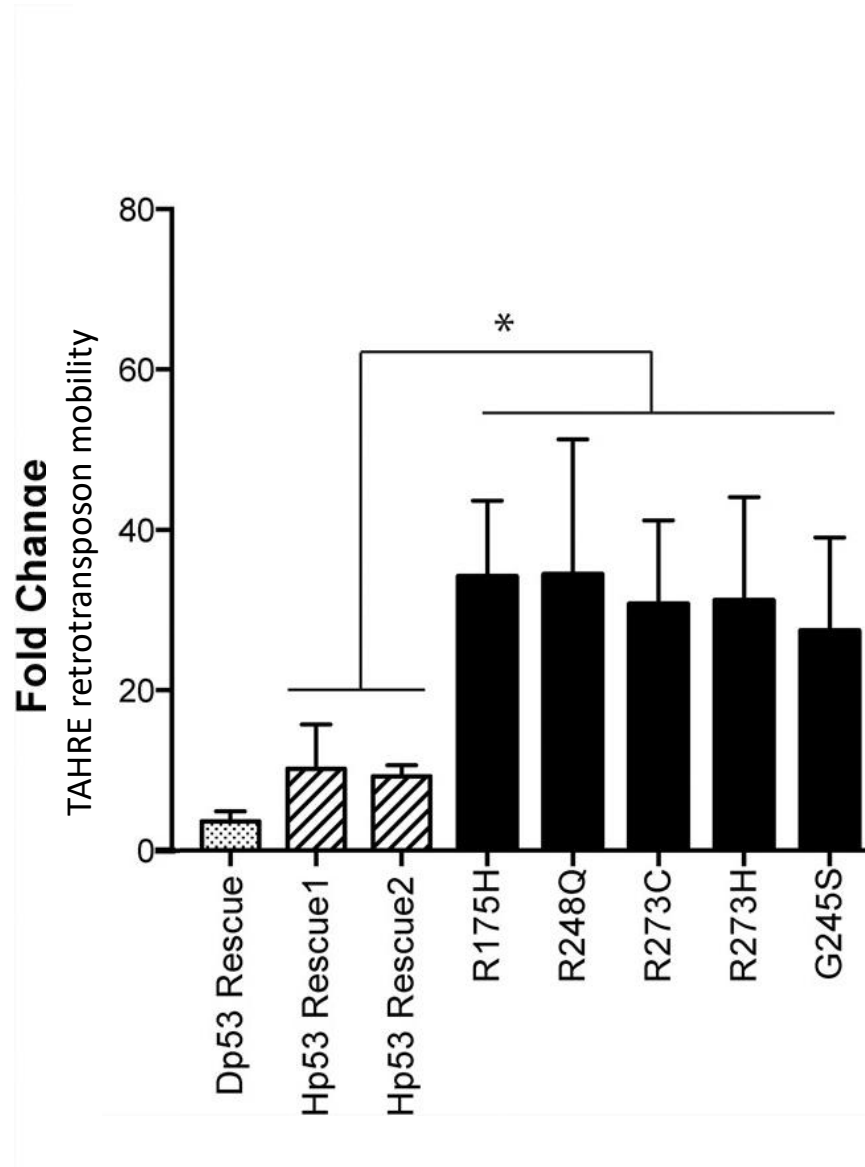


1. WT : wild type fly
2. p53^{-/-} : P53 mutant fly
3. Dp53 Rescue: p53 mutant fly + transgenic wild type *Drosophila* p53
4. Hp53 Rescue1: p53 mutant fly + transgenic wild type human p53
5. Hp53 Rescue2: p53 mutant fly + transgenic wild type human p53 (replicate)
6. Black bars: p53 mutant fly + a transgenic human mutant p53 allele from a cancer patient

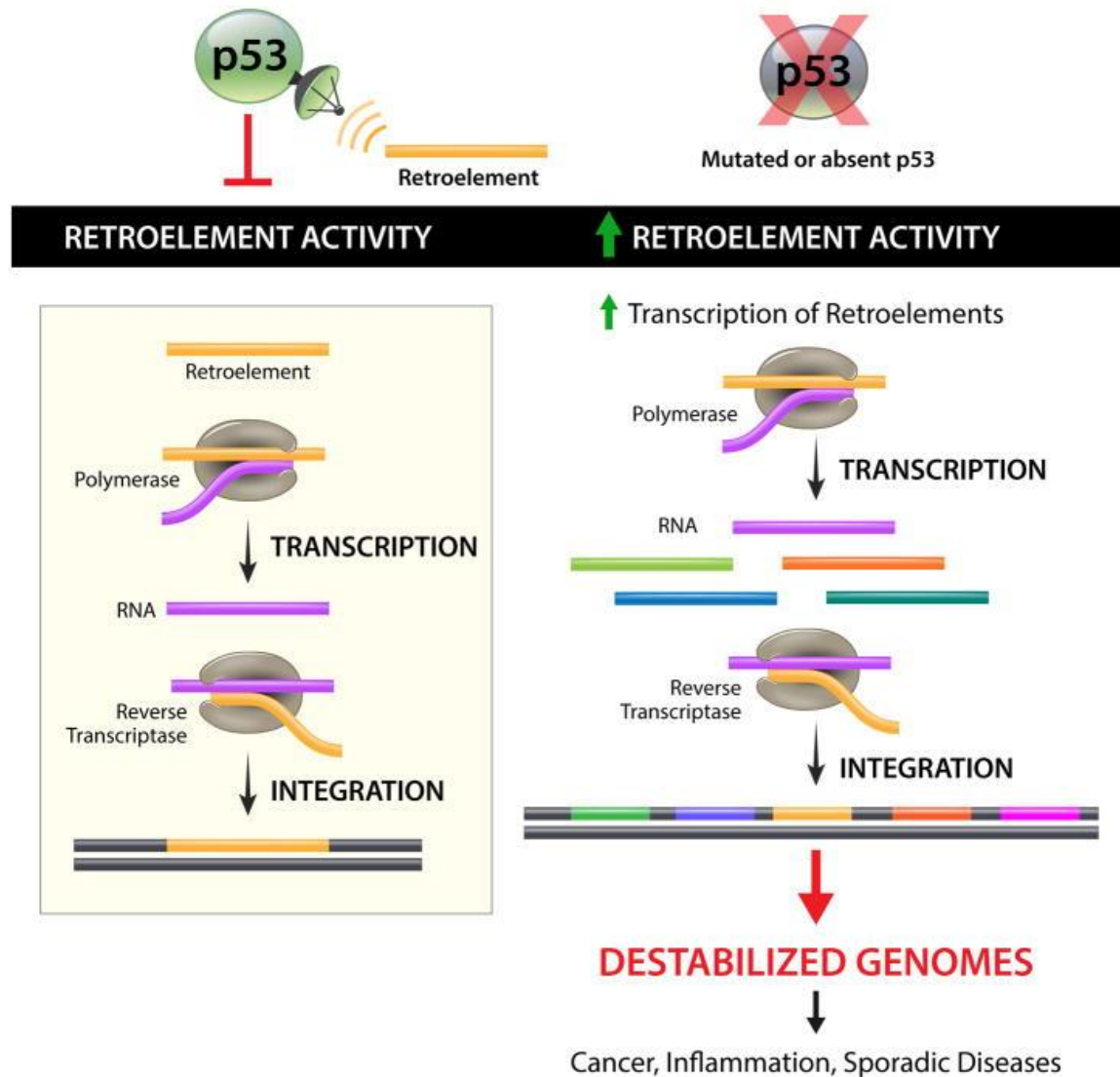
Think Break

Data Interpretation

Which strain has the most p53 activity?



The p53 protein promotes genome stability by silencing transposon mobility



DNA Mutations

Causes and Consequences

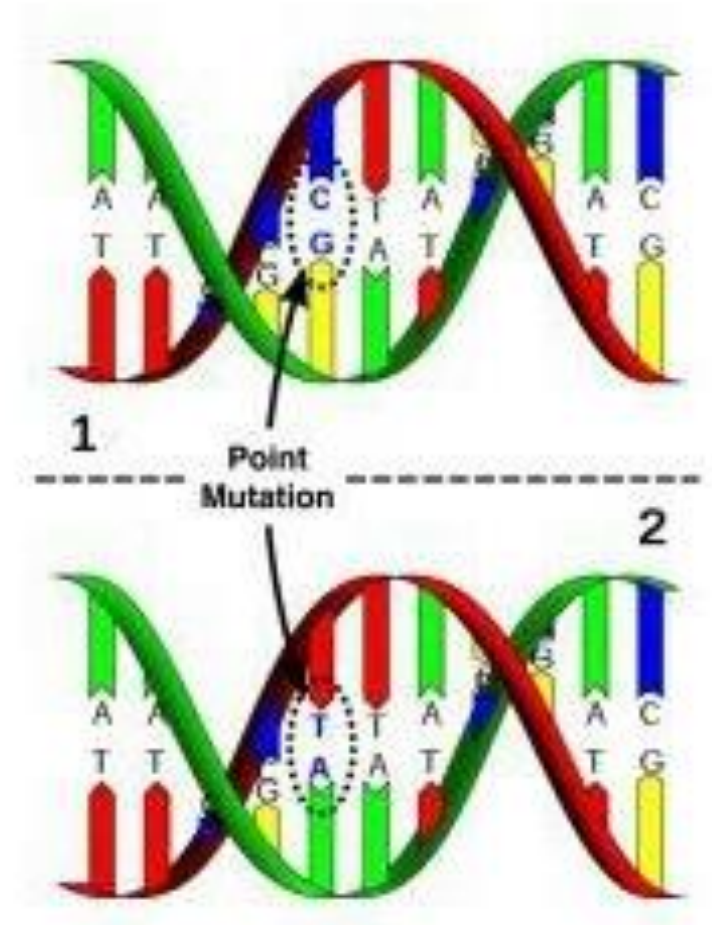
What is a mutation?

What are the consequences of mutations?

Where do mutations come from?

A mutation is a change in DNA sequence

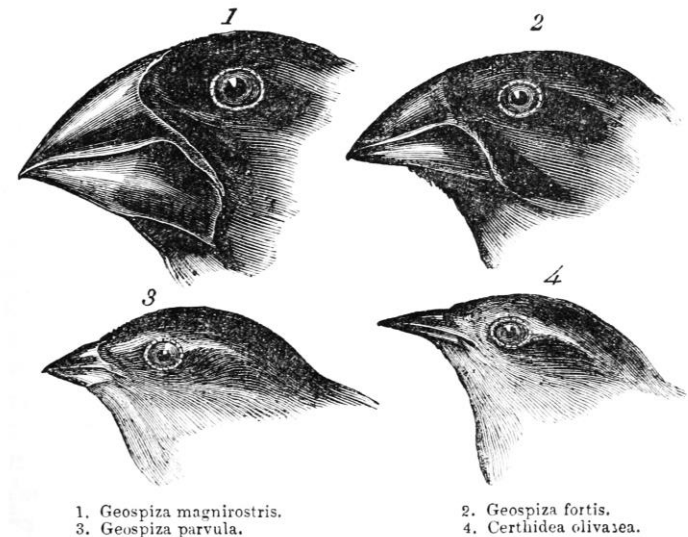
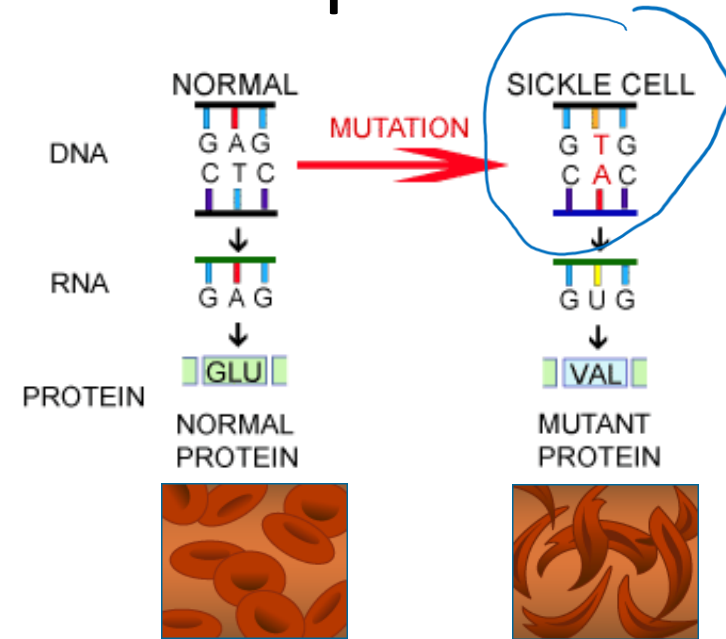
- Any unpredictable change in the structure or amount of the DNA of an organism is called a **mutation**.
- Most mutations occur in **somatic (body) cells** and are not passed on from one generation to the next.
- Only those mutations that occur in the formation of **gametes** can be inherited.



A mutation is a change in DNA sequence

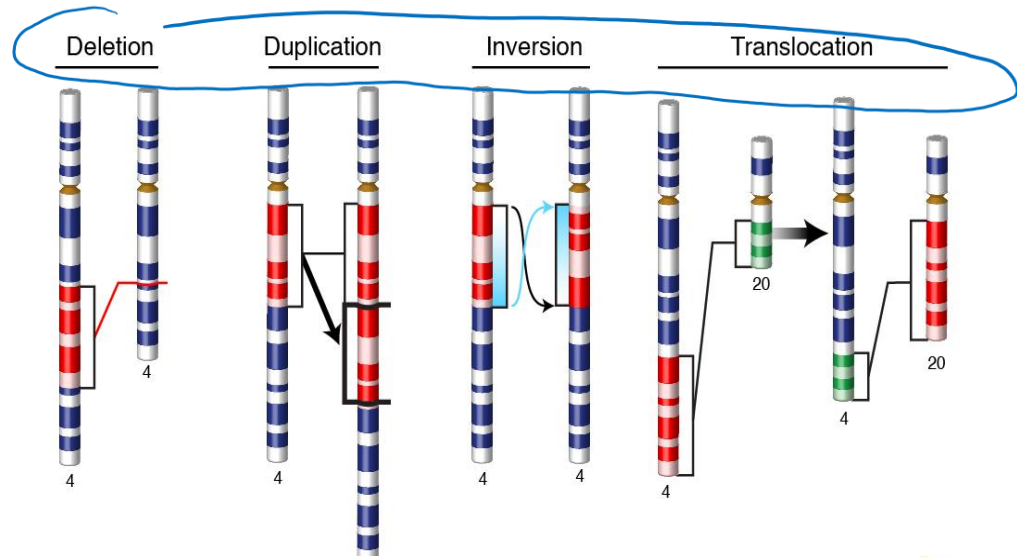
Mutations can:

- be detrimental, neutral or beneficial
- affect coding sequences or regulatory sequences
- be an important source of genetic variation
 - Germline mutations are heritable.
 - Recombination during meiosis generates different combinations of alleles
 - This allows for adaptation/evolution

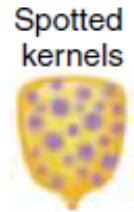
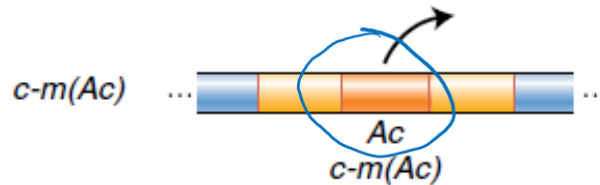


Mutations can occur on different scales

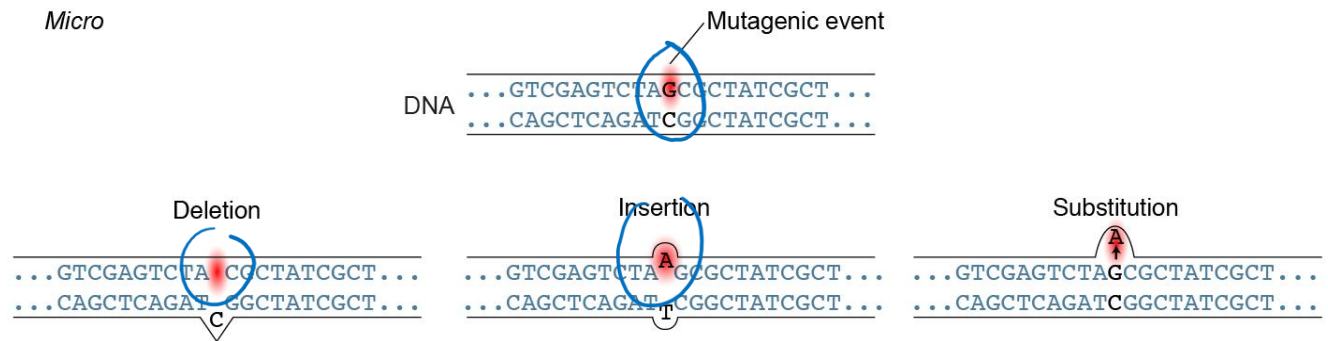
Chromosomal mutations: gain or loss of all or part of a chromosome



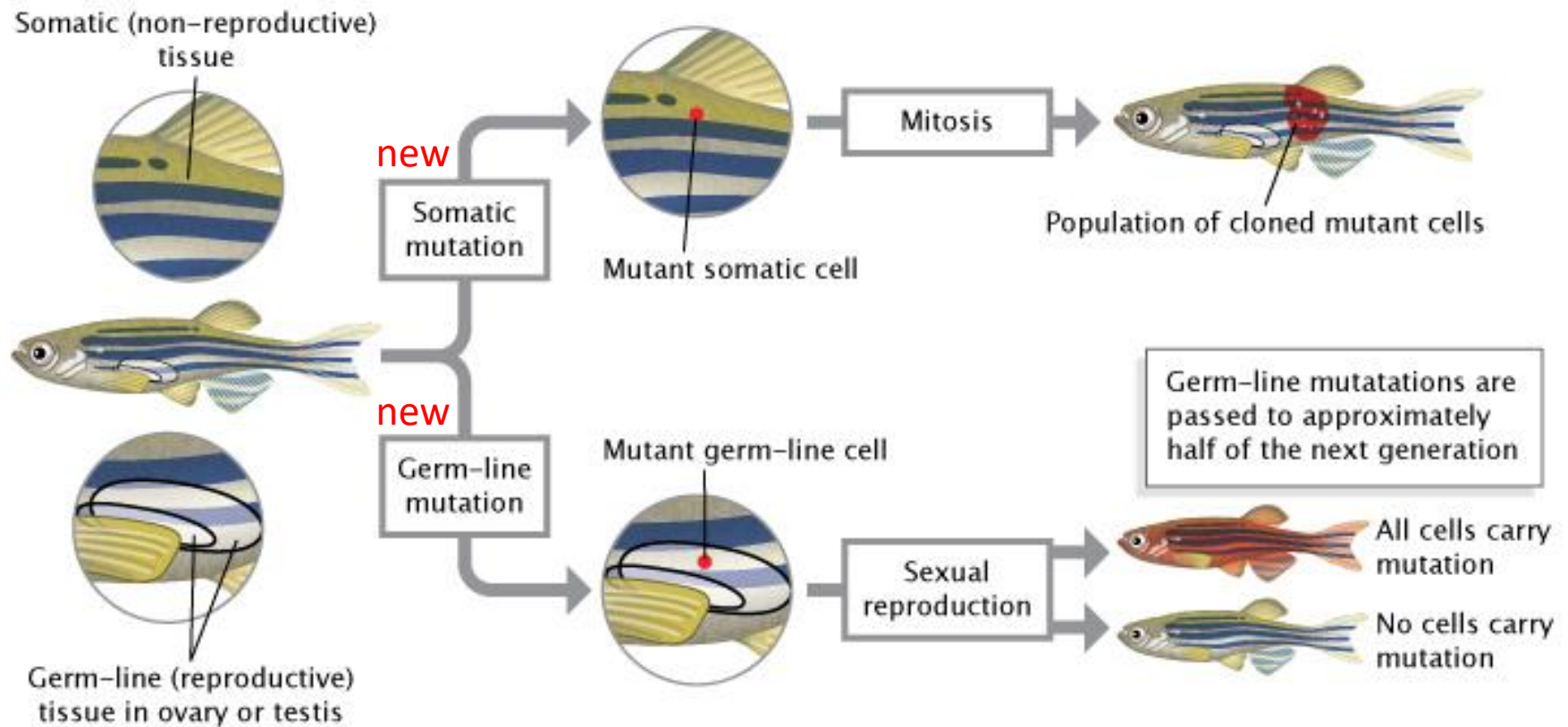
Insertional mutations: insertion of large regions of DNA, e.g. transposable elements



Point mutations: changes in a single nucleotide or addition/deletion of one or more nucleotides

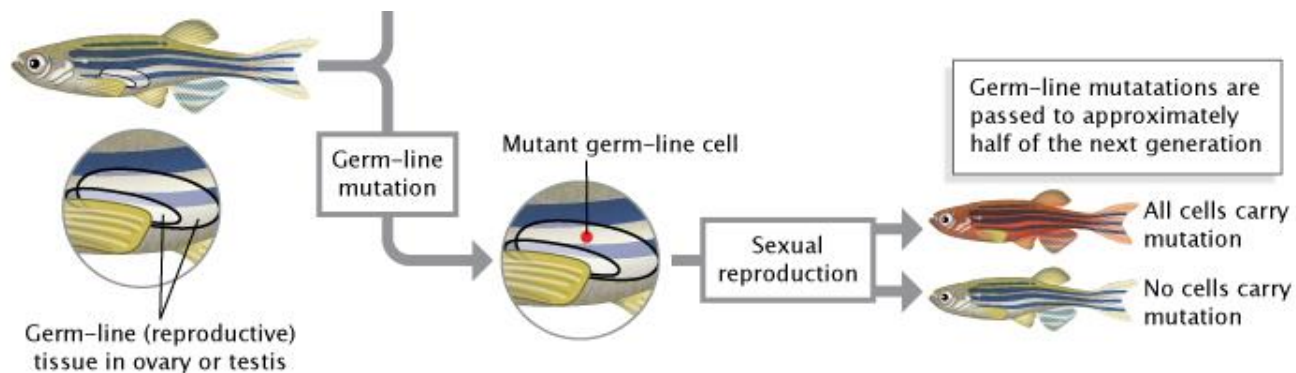


Mutations in the germline, but not the soma, can be inherited



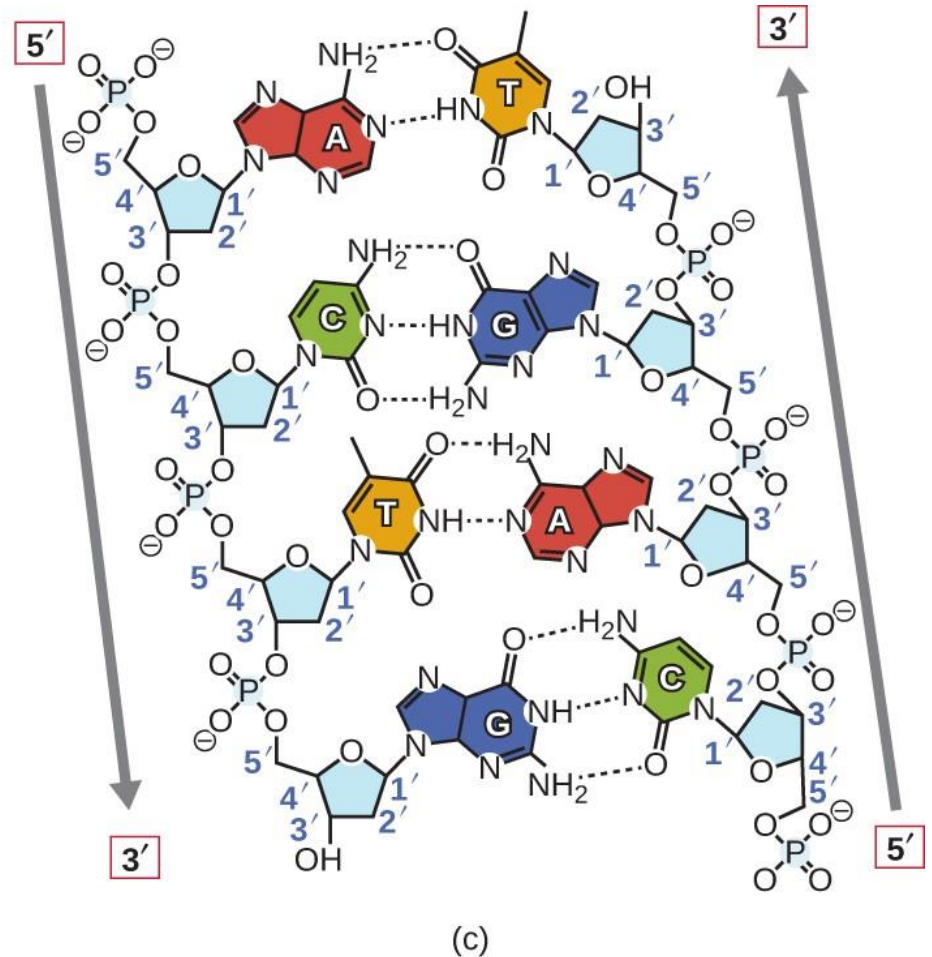
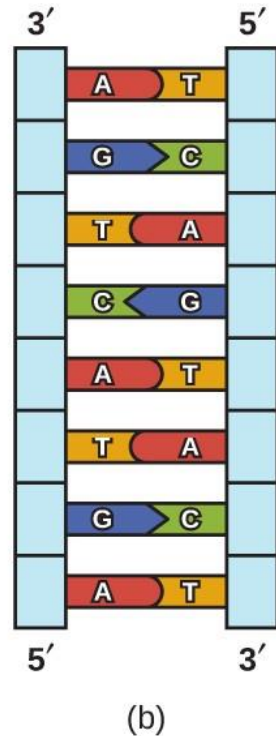
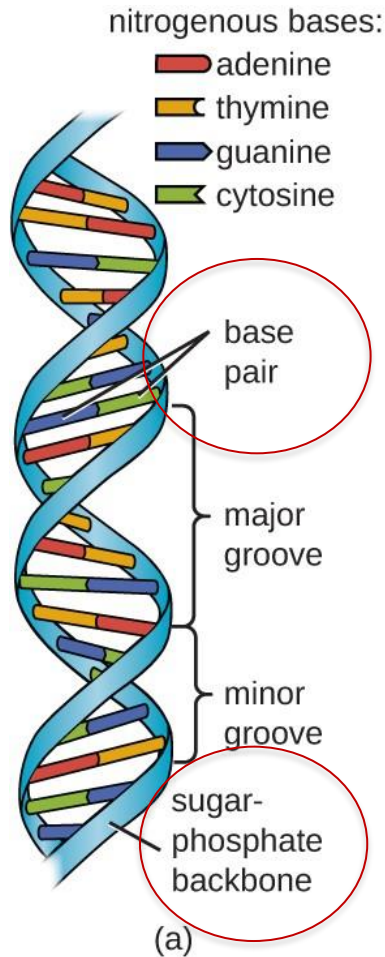
Think Break

Why will only half of the F1 individuals carry the mutation?

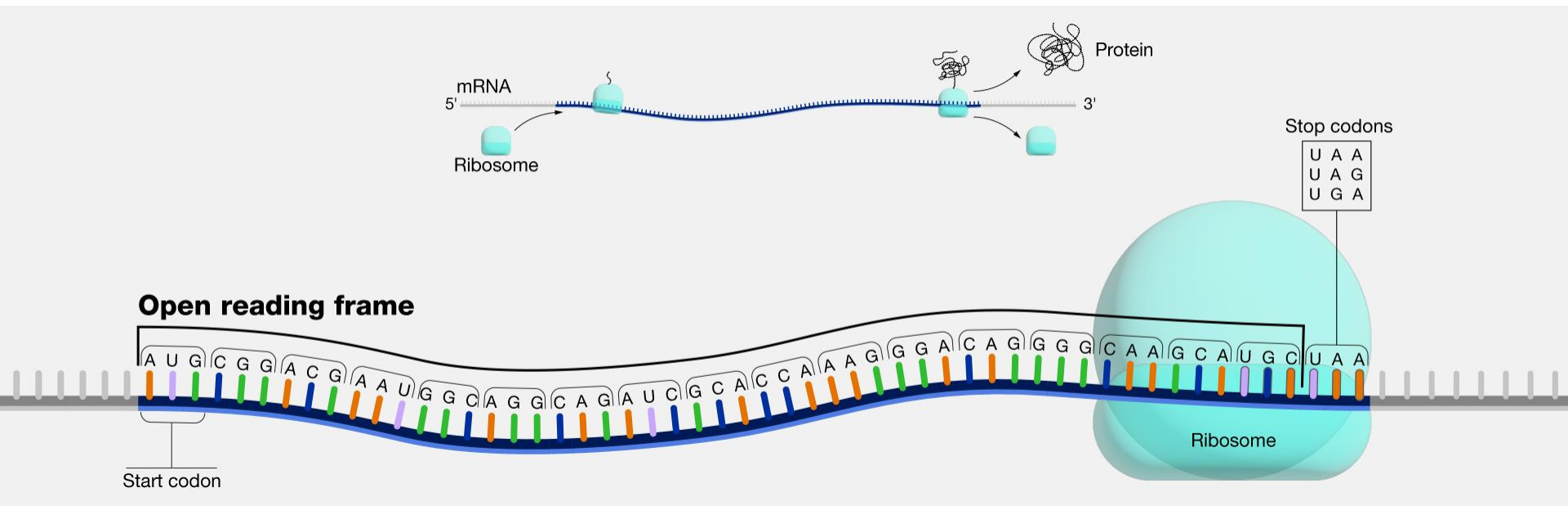


Quick DNA Refresher:

DNA structure



Quick DNA Refresher: open reading frames (ORFs)



Quick Refresher: The Genetic Code

Amino Acids			Codons						
Alanine	Ala	A	GCA	GCC	GCG	GCU			
Cysteine	Cys	C	UGC	UGU					
Aspartic acid	Asp	D	GAC	GAU	GAC	GAU			
Glutamic acid	Glu	E	GAA	GAG					
Phenylalanine	Phe	F	UUC	UUU					
Glycine	Gly	G	GGA	GCG	GGG	GGU			
Histidine	His	H	CAC	CAU					
Isoleucine	Ile	I	AUA	AUC	AUU				
Lysine	Lys	K	AAA	AAG					
Leucine	Leu	L	UUA	UUG	CUA	CUC	CUG	GUU	
Methionine	Met	M	AUG						
Asparagine	Asn	N	AAC	AAU					
Proline	Pro	P	CCA	CCC	CCG	CCU			
Glutamine	Gln	Q	CAA	CAG					
Arginine	Arg	R	AGA	AGG	CGA	CGC	CGG	CGU	
Serine	Ser	S	AGC	AGU	UCA	UCC	UCG	UCU	
Threonine	Thr	T	ACA	ACC	ACG	ACU			
Valine	Val	V	GUA	GUC	GUG	GUU			
Tryptophan	Trp	W	UGG						
Tyrosine	Tyr	Y	UAC	UAU					

Categories of point mutations

Base substitutions

Transition

purine → purine

pyrimidine → pyrimidine

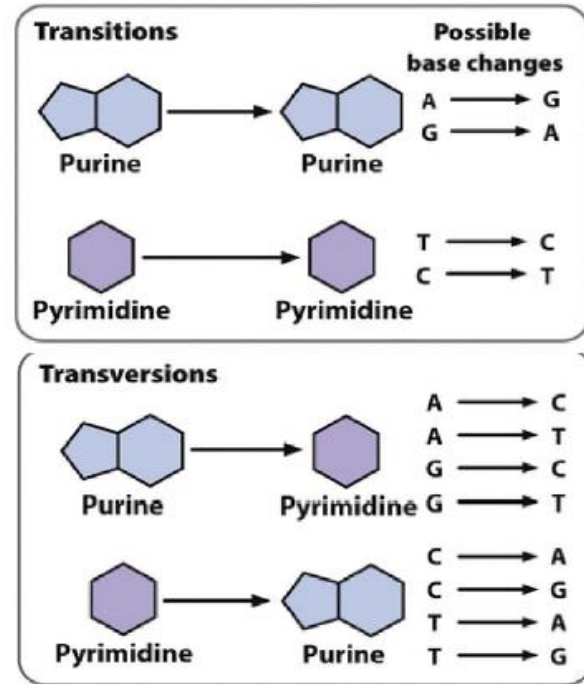
e.g. GC→AT

Transversion

pyrimidine → purine

purine → pyrimidine

e.g. CG→TA



Base additions and deletions (indels)

AAGACTCCT → AAGA**G**CTCCT
AA**G**ACTCCT → AA**A**CTCCT

Categories of point mutations in open reading frames

Synonymous mutations (also called silent mutations)

change the sequence but not the amino acid it encodes

Nonsynonymous mutations (also called missense mutations)

change the sequence of a codon to one that codes for a different amino acid.

Conservative mutations: mutant codon encodes a chemically similar amino acid
(e.g. lys-to-arg)

Non-conservative mutations: mutant codon encodes a chemically different amino acid
(e.g. lys-to-thr)

Nonsense mutations

change the sequence of a codon that codes for an amino acid into one that stops translation
(i.e. UAA, UAG, or UGA).

Effect depends on distance from the 3' end of the ORF

Can trigger nonsense-mediated decay (NMD), which completely degrades the mRNA

Frameshift mutations

Caused by an insertion or deletion

Change the translation reading frame for all codons downstream of the mutation

Consequences of point mutations in open reading frames

		Single base substitution									
		No mutation	Silent mutation (synonymous)	Missense mutation (nonsynonymous)				Nonsense mutation			
				Conservative		Nonconservative					
DNA	CAT AAG CAG AGT ACT GTA TTC GTC TCA TGA	CAT AA A CAG AGT ACT GTA TTT GTC TCA TGA	CAT A GG CAG AGT ACT GTA T CC GTC TCA TGA	CAT A CG CAG AGT ACT GTA T GC GTC TCA TGA	CAT T AG CAG AGT ACT GTA A TC GTC TCA TGA						
mRNA	CAU AAG CAG AGU ACU	CAU AA A CAG AGU ACU	CAU A GG CAG AGU ACU	CAU A CG CAG AGU ACU	CAU U AG CAG AGU ACU						
Protein	His Lys Gln Ser Thr	His Lys Gln Ser Thr	His Arg Gln Ser Thr	His Thr Gln Ser Thr	His Stop						

Consequences of point mutations in open reading frames

		Single base insertion or base deletion (indel)	
No mutation		Insertion (frameshift)	Deletion (frameshift)
DNA	CAT TGC GAC AAG GAT AGT ACT CCT GTA ACG CTG TTC CTA TCA TGA GGA	CAT G TG CGA CAA GGA TAG TAC TCC T ↓ GTA C AC GCT GTT CCT ATC ATG AGG A ↑	CAT T GCG ACA AGG ATA GTA CTC CT ↑ GTA A CGC TGT TCC TAT CAT GAG GA ↓
mRNA	CAU UGC GAC AAG GAU AGU ACU CCU	CAU GUG CGA CAA GGA UAG UAC UCC U	CAU GCG ACA AGG AUA GUA CUC CU
Protein	His Cys Asp Lys Val Ser Thr Pro	His Val Arg Gln Gly Stop	His Ala Thr Arg Ile Val Leu

Point mutations can have different effects on gene function, depending on their location

- Wild type function (mutation has no effect on gene function)
- Loss-of-function (gene function is partially or completely lost)
 - HYPOMORPHIC: Weak/partial – protein retains some function or is produced at a reduced level
 - NULL: Protein is non-functional or not produced
- Gain-of-function (gene function is increased or novel)
 - HYPERMORPHIC: Protein is hyperactive
 - ECTOPIC: More protein made or made in wrong time/place
 - NEOMORPHIC: Protein gains new function

Mutations can be in coding or non-coding regions

Example: loss-of-function allele

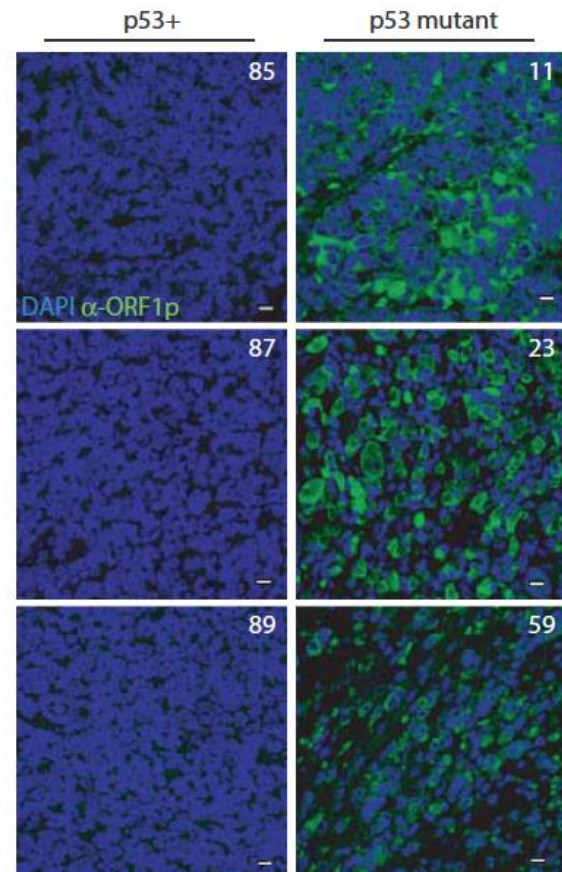
Mutant p53 alleles associated with human cancers typically contain mutations that reduce or eliminate their function.

Element	Transposition	Structure
LINEs	Autonomous	
SINEs	Nonautonomous	
DNA transposons	Autonomous	
	Nonautonomous	

Deregulated retroelements stratify with p53 mutations in Wilms tumors. Compared with Wilms tumors that are wild type for p53 (*left* panels: 85, 87, and 89), Wilms tumors that are mutant for p53 (*right* panels: 11, 23, and 59) show dramatically elevated human LINE-1 ORF1p expression (α -ORF1p [green]; counterstained to visualize DNA [blue])

[p53 genes function to restrain mobile elements - PubMed \(nih.gov\)](#)

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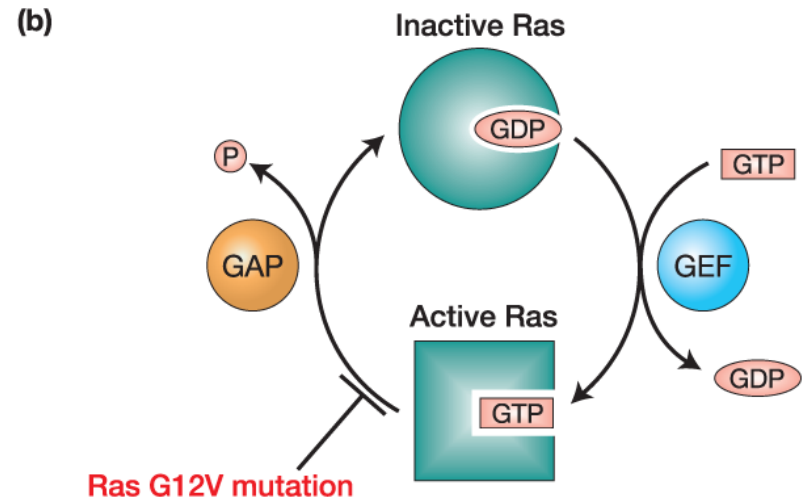
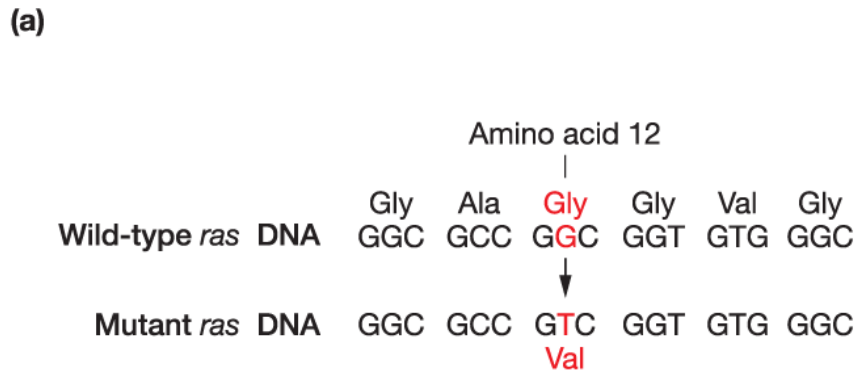


blue = DNA
green = human LINE-1 ORF1p expression

Example: gain-of-function allele

Mutant alleles of the *ras* gene that associated with human cancers typically contain mutations that make them constitutively active.

A point mutation in the *ras* ORF affects Ras protein function



Effect on the gene

Effect on the gene product

Effect on the phenotype

- The uncontrolled cell proliferation resulting from the mutant Ras protein can lead to cancer.
- Approximately 30% of human cancers are driven by mutations in Ras.

Sequences in non-coding regions can be important for gene function

Point mutations can alter mRNA splicing*

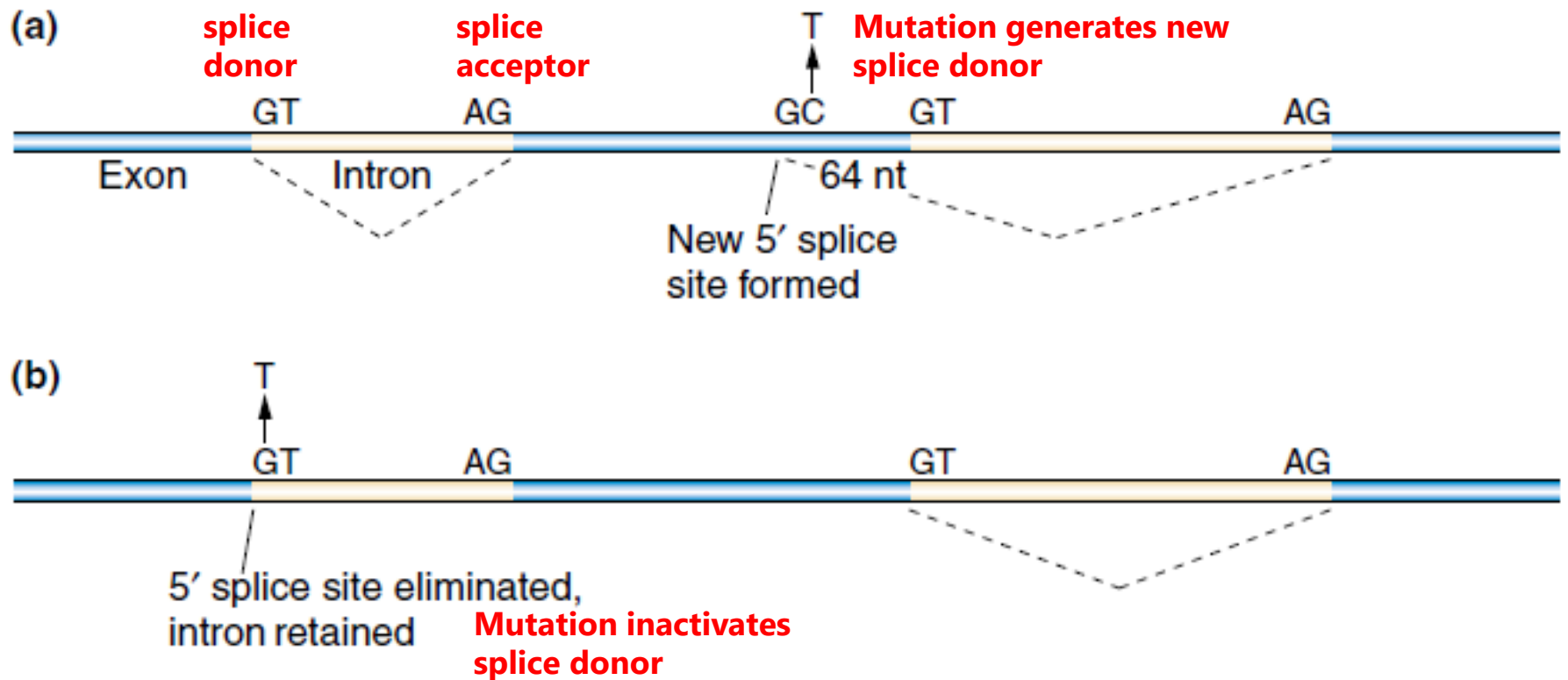
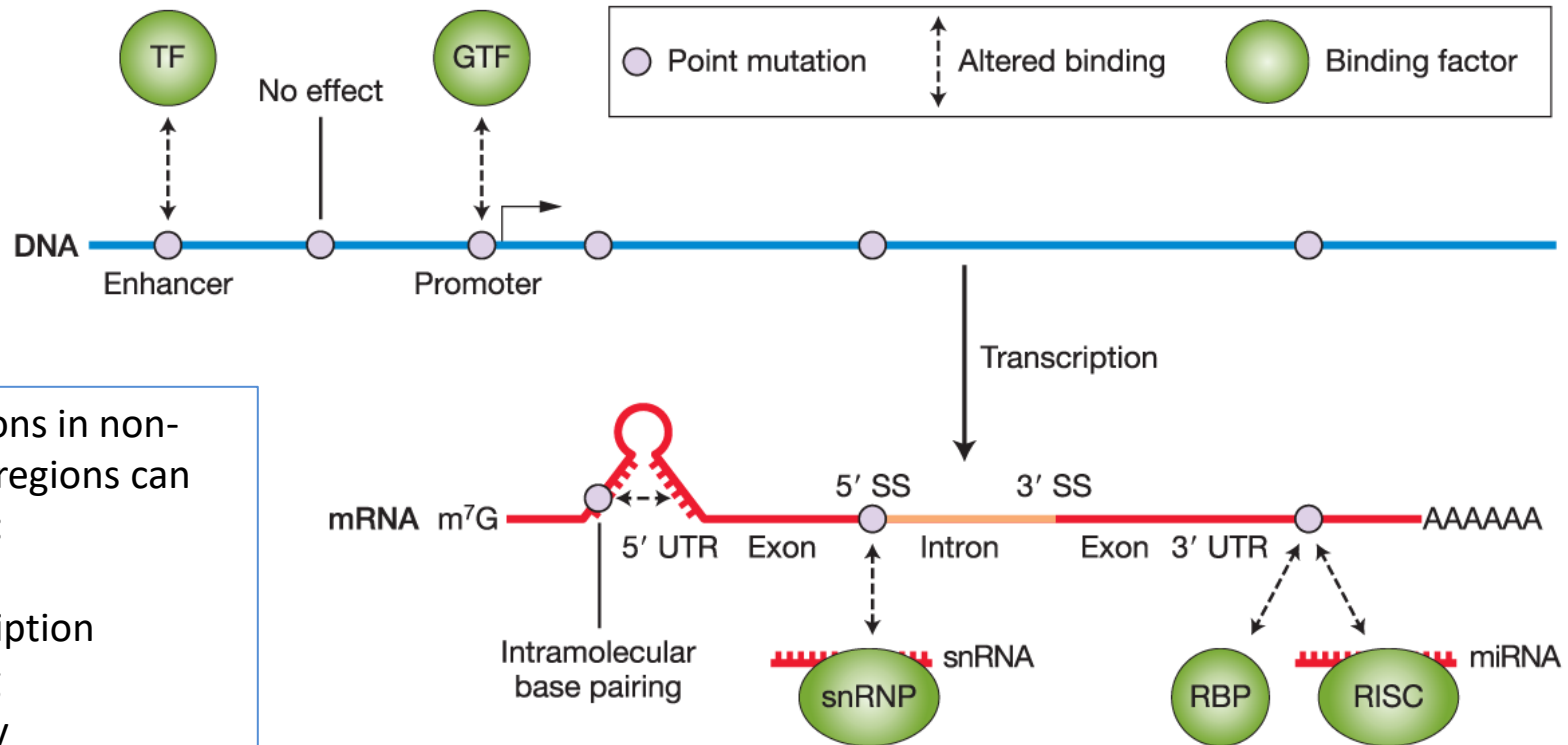


Figure 16-3
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* can be in non-coding or coding regions

Sequences in non-coding regions can be important for gene function

Consequences of point mutations in noncoding regions

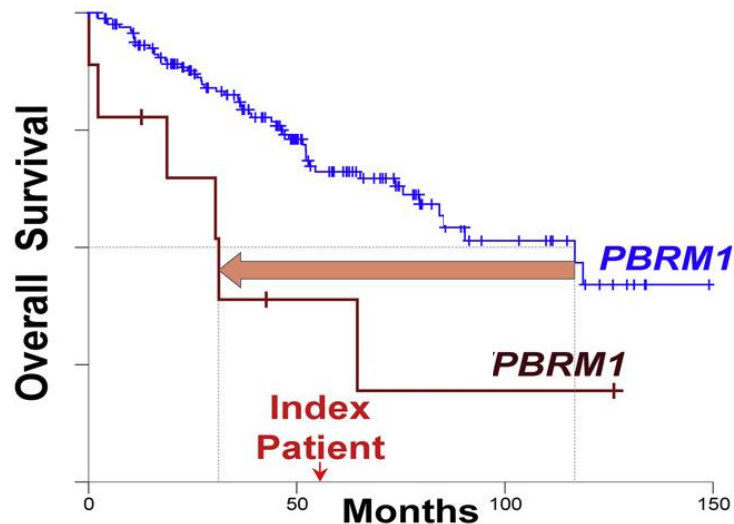


Mutations in non-coding regions can disrupt:

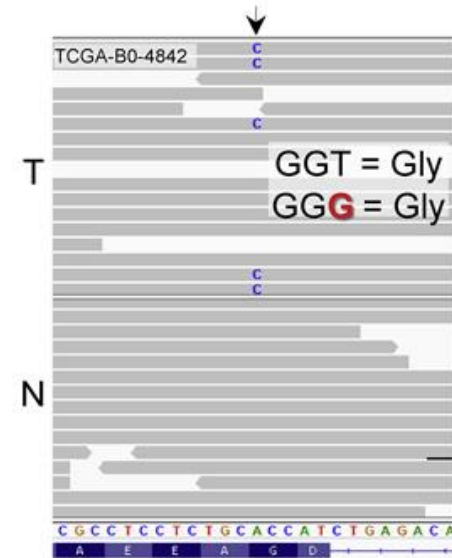
Transcription
Splicing
Stability
Translation
Function
etc.

Example: A synonymous mutation can cause a loss of gene function, though rarely.

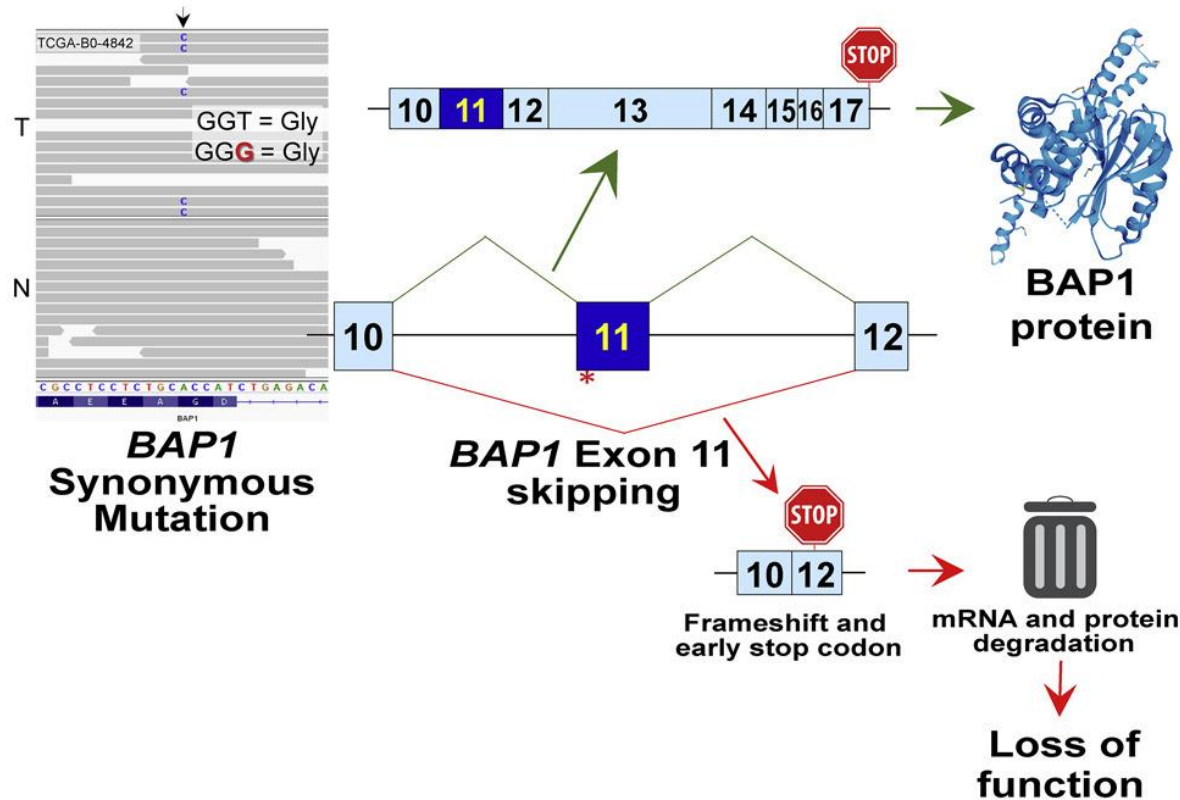
Observation: A kidney cancer patient with a mutation in the PBRM1 gene had a much worse outcome than expected. Why?



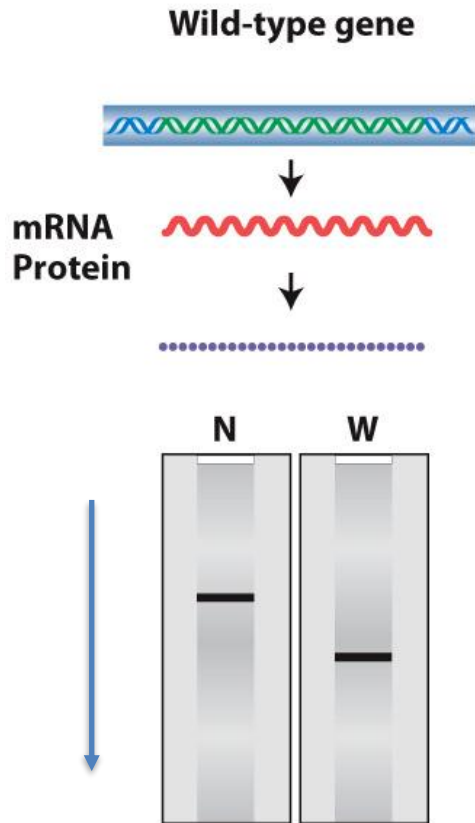
Hypothesis: This patient has a mutation in another gene, in addition to PBRM1, that makes the cancer outcome worse.



Example: A synonymous mutation in a BAP1 exon leads to loss of BAP1 function through exon skipping.



Detecting effects of point mutations on a gene's mRNA and protein products



Sometimes, but not always, mutations lead to detectable changes in the mRNA or protein generated from the gene.

Reminders:

N: Northern blots are for looking at RNA.

W: Western blots are for looking at protein.

Electrophoretic mobility (position on gel) reflects size, band intensity reflects amount present.

Consequences of point mutations within genes

NOTE: an early stop codon can lead to mRNA degradation (via “nonsense mediated decay”) which can lead to no RNA and protein being produced at all.

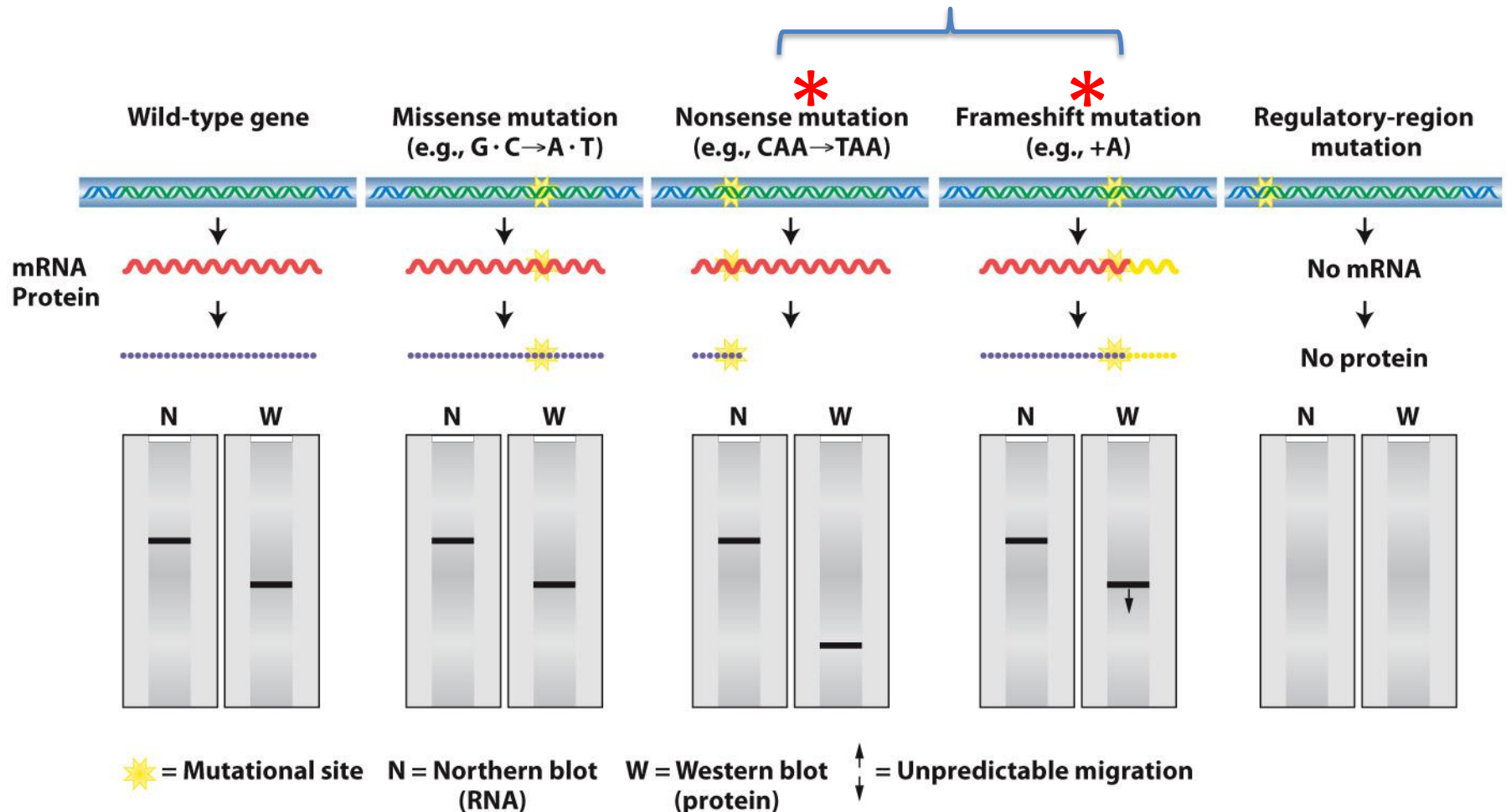


Figure 16-4
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Mutations can be spontaneous or induced

spontaneous: occur naturally, arise in all cells

induced: caused by exposure to ionizing radiation, mutagenic chemicals, etc., which increase the rate at which mutations occur

